

Fig. 3: Actual-size PCB pattern of the RGB colour generator

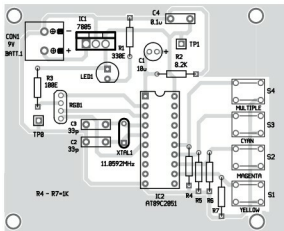


Fig. 4: Component layout of the PCB

IC2 is used to drive common-anode RGB LED, with P3.0 connected to red pin, P3.1 to green pin and P3.2 to blue pin of the RGB LED. Port1 is connected to the four tactile switches S1 through S4 for producing multiple colours.

When the circuit is switched on, red colour of the RGB LED glows for a preprogrammed time interval of three seconds. It then switches off for three seconds. This

EFY Note

is followed by green and blue colours switching on and off in a similar manner and for the same time periods.

Now, when we press switch S1, we get yellow colour (red + green), and if we press switch S2, we get magenta (red + blue). If we press switch S3, we get cyan (blue + green), and if we press switch S4, we get multiple colours with the help of PWM technique.

Software

The software is written in embedded C language and compiled using Keilvision 4 version compiler. It contains simple switch statements to produce different colours from primary colours. We have used Topwin 6 software to burn the hex code into the MCU using a Universal Topwin programmer board.

Construction and testing

An actual-size, single-side PCB of the circuit is shown in Fig. 3 and its component layout in Fig. 4. Assemble the circuit on the PCB as it minimises time and assembly errors. Carefully assemble the components and double-check for any error(s). Use a proper IC base for the MCU.

CON1 is a 2-pin connector used to connect a 9V battery.

For troubleshooting, verify the voltages listed in the test points table. **EPY**



Control with Precision



RE57

Temperature Control

- Universal Measuring Input
- Digital Input Filter - Timer
settable from 0 ... 999.9 seconds
- Control Type: PID and ON / OFF
- Onsite Mode Selection: Auto / Manual
- Timer Function: with 5 functional selection
- Onsite programmability - Heating / cooling
- Output on Failure function



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